Exercise C - II

1. If the first 3 consecutive terms of a geometrical progression are the real roots of the equation $2x^3 - 19x^2 + 57x - 54 = 0$ find the sum to infinite number of terms of G.P.

यदि एक गुणोतर श्रेणी के प्रथम तीन क्रमागत पद समीकरण $2x^3 - 19x^2 + 57x - 54 = 0$ के वास्तविक मूल है। तो गु. श्रे. के अनन्त पदों का योग निकालिये।

Ans. $\frac{27}{2}$

Sol. $2x^3 - 19x^2 + 57x - 54 = 0$ (α, β, γ are roots)

$$\alpha+\beta+\gamma = \frac{19}{2},\ \alpha\beta+\beta\gamma+\gamma\alpha = \frac{57}{2}, \alpha\beta\gamma = \frac{54}{2}$$

α,β,γ are in G.P.

$$\frac{a}{r}, a, ar = \alpha.\beta, \gamma$$

$\alpha\beta\gamma = a^3 = 27 \Rightarrow a = 3$

$$\frac{a}{r} + ar + a = \frac{19}{2}$$

$$3\left(\frac{1}{r}+1+r\right) = \frac{19}{2}$$

$6[1+ r^2 + r] = 19r$

$6r^2 - 13r + 6 = 0$

$6r^2 - 9r - 4r + 6 = 0$

$3r(2r - 3) - 2(2r - 3) = 0$

$(3r - 2)(2r - 3) = 0$

$r = \frac{2}{3}, r = \frac{3}{2}$ not possible

$$S_\infty = \frac{\frac{a}{r}}{1-r} = \frac{\frac{3}{\left(\frac{2}{3}\right)}}{1-\frac{2}{3}} = \frac{27}{2}$$

2. Find the sum of the infinite series $\frac{1.3}{2} + \frac{3.5}{2^2} + \frac{5.7}{2^3} + \frac{7.9}{2^4} + ...\infty$.

अनन्त श्रेणी $\frac{1.3}{2} + \frac{3.5}{2^2} + \frac{5.7}{2^3} + \frac{7.9}{2^4} + ...\infty$ का योग निकालिए।

Ans. 23

Sol. $S = \frac{1.3}{2} + \frac{3.5}{2^2} + \frac{5.7}{2^3} + \frac{7.9}{2^4} \ldots\ldots\ldots\infty$ ..... (i)

$S\left(\frac{1}{2}\right) = \frac{1.3}{2^2} + \frac{3.5}{2^3} + \frac{5.7}{2^4} + \ldots\ldots\infty$ ..... (ii)

(i) – (ii)

$\frac{S}{2} = \frac{1.3}{2} + \frac{3.4}{2^2} + \frac{4.5}{2^3} + \frac{4.7}{2^4} \ldots\ldots\ldots\infty$ .....

$\frac{S}{2} = \frac{1\cdot3}{2} + 4\left[\frac{3}{2^2} + \frac{5}{2^3} + \frac{7}{2^4} + \ldots\ldots\ldots\right]$

$\frac{S}{2} = \frac{1\cdot3}{2} + \left[3 + \frac{5}{2} + \frac{7}{2^2} + \ldots\ldots\right]$

Let $S' = 3 + \frac{5}{2} + \frac{7}{2^2} + \ldots\ldots\ldots$ ...(iii)

$\frac{S'}{2} = 0 + \frac{3}{2} + \frac{5}{2^2} + \ldots\ldots\ldots$ ...(iv)

(iii) – (iv)

$\frac{S'}{2} = 3 + \left[\frac{2}{2} + \frac{2}{2^2} + \ldots\ldots\right]$

$= 3 + \left[1 + \frac{1}{2} + \ldots\ldots\right]$

$\frac{S'}{2} = 5 \Rightarrow S' = 10$

By equation (ii)

$\frac{S}{2} = \frac{3}{2} + 10 = \frac{23}{2}$

S = 23

3. One of the roots of the equation $2000x^6 + 100x^5 + 10x^3 + x - 2 = 0$ is of the form $\frac{m+\sqrt{n}}{r}$, where m is non zero integer and n and r are relatively prime natural numbers. Find the value of m + n + r.

समीकरण $2000x^6 + 100x^5 + 10x^3 + x - 2 = 0$ का एक मूल $\frac{m+\sqrt{n}}{r}$ के रूप में है जहाँ m अशून्य पूर्णांक तथा n एवं r सापेक्षतः अभाज्य प्राकृतिक संख्यायें है। m + n + r का मान निकालिए।

Ans. 200

Sol. $2000x^6 + 100x^5 + 10x^3 + x - 2 = 0$

$$\alpha = \frac{m+\sqrt{n}}{r} \Rightarrow \beta = \frac{m-\sqrt{n}}{r}$$

$x + 10x^3 + 100x^5 = 2 - 2000x^6$

$$\frac{x\left(\left(10x^2\right)^3 - 1\right)}{10x^2 - 1} = 2\left(1 - \left(10x^2\right)^3\right)$$

$$\frac{x\left(1 - \left(10x^2\right)^3\right)}{1 - 10x^2} = 2\left(1 - \left(10x^2\right)^3\right)$$

$$\left[1 - \left(10x^2\right)^3\right]\left[\frac{x}{1-10x^2} - 2\right] = 0$$

$1000x^6 - 1 = 0$ or $x - 2 + 20x^2 = 0$

$x^6 = \frac{1}{1000}$ or $20x^2 + x - 2 = 0$

$$x = \frac{-1 \pm \sqrt{161}}{40}$$

$\Rightarrow m = -1, n = 161, = 40$

$m + n + r = 200$

4. Let $S = \sum_{n=1}^{99} \frac{5^{100}}{(25)^n + 5^{100}}$. Find [S]. Where [y] denotes largest integer less than or equal to y.

माना कि $S = \sum_{n=1}^{99} \frac{5^{100}}{(25)^n + 5^{100}}$ तो [S] का मान बताओं। जहाँ [y] महत्तम पूर्णांक फलन है।

Ans. 49

Sol. $$S = \sum_{n=1}^{99} \frac{5^{100}}{(25)^n + 5^{100}}$$

$$S = \sum_{n=1}^{99} \frac{1}{1 + 5^{2n-100}}$$

$$= \frac{1}{1+5^{-98}} + \frac{1}{1+5^{-96}} + \frac{1}{1+5^{-94}} + \ldots\ldots\ldots + \frac{1}{1+5^{-2}} + \frac{1}{1+5^{0}} + \frac{1}{1+5^{2}} + \ldots\ldots + \frac{1}{1+5^{98}}$$

$$=\left(\frac{1}{1+5^{-98}}+\frac{1}{1+5^{98}}\right)+\left(\frac{1}{1+5^{-96}}+\frac{1}{1+5^{96}}\right)+.....+\left(\frac{1}{1+5^{-2}}+\frac{1}{1+5^{2}}\right)+\frac{1}{1+50}$$

$$S=\underbrace{1+1+.....+1}_{49\text{ times}}+\frac{1}{2}$$

$S = 49\cdot5$

$[S] = [49\cdot5] = 49$

5. If $3^{2\sin 2x-1}$, 14, $3^{4-2\sin 2x}$ form first three terms of an A.P., then find the sum $1 + \sin 2x + \sin^2 2x + ... \infty$.

यदि $3^{2\sin 2x-1}$, 14, $3^{4-2\sin 2x}$ किसी समान्तर श्रेणी के प्रथम तीन पद है तो $1 + \sin 2x + \sin^2 2x + ... \infty$ का मान बताओं।

Ans. 2

Sol. $3^{2\sin 2x-1}$, 14, $3^{4-2\sin 2x}$ are in A.P.

$2 \times 14 = 3^{2\sin 2x-1} + 3^{4-2\sin 2x}$

$$28=\frac{3^{2\sin 2x}}{3}+\frac{3^4}{3^{2\sin 2x}}$$

$$28=\frac{t}{3}+\frac{81}{t}$$

$$28=\frac{t^2+243}{3t}$$

$t^2 - 84t + 243 = 0$

$t^2 - 81t - 3t + 243 = 0$

$t(t - 81) - 3(t - 81) = 0$

$(t - 3)(t - 81) = 0$

$t = 3, t = 81$

$3^{2\sin 2x} = 3^1$	or	$3^{2\sin 2x} = 3^4$
$2\sin 2x = 1$	or	$2\sin 2x = 4$
$\sin 2x = \frac{1}{2}$		$\sin 2x = 2$
		Not possible

$S = 1 + \sin 2x + \sin^2 2x + ...$

$$S=1+\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+.....=\frac{1}{1-\frac{1}{2}}=2$$

6. Given that the cubic $ax^3 - ax^2 + 9bx - b = 0$ $(a \neq 0)$ has all three positive roots. Find the harmonic mean of the roots independent of a and b, hence deduce that the root are all equal. Find also the minimum value of $(a + b)$ if a and $b \in N$.

घन समीकरण $ax^3 - ax^2 + 9bx - b = 0$ $(a \neq 0)$ के तीनों मूल धनात्मक है। a एवं b का हरात्मक माध्य निकालिए तदानुसार बताइए कि सभी मूल, समान है। (a + b) का न्युनतम मान भी निकालिए यदि $a, b \in N$ हो।

Ans. 28

Sol. $ax^3 - ax^2 + 9bx - b = 0$ $(a \infty 0)$
Let α, β, γ are roots.

$$H = \frac{3}{\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}} = \frac{3\alpha\beta\gamma}{\alpha\beta+\beta\gamma+\gamma\alpha} = \frac{3\left(\frac{b}{a}\right)}{\left(\frac{ab}{a}\right)} = \frac{1}{3}$$

Applying A.M., G.M., H.M.

$$\frac{\alpha+\beta+\gamma}{3} \geq (\alpha\beta\gamma)^{\frac{1}{3}} \geq \frac{3}{\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}}$$

$$\frac{1}{3} \geq (\alpha\beta\gamma)^{\frac{1}{3}} \geq \frac{1}{3}$$

$\Rightarrow$ A.M. = G.M. = H.M.
$\Rightarrow \alpha = \beta = \gamma$

$\because\ \alpha + \beta + \gamma = 1 \Rightarrow \alpha = \beta = \gamma = \frac{1}{3}$

$$\left(x - \frac{1}{3}\right)^3 = x^3 - x^2 + \frac{x}{3} - \frac{1}{27} = 0$$

$27x^3 - 27x^2 + 9x - 1 = 0$
compare with (1)
$\Rightarrow a = 27, b = 1 \qquad \Rightarrow a + b = 28$

7. The sequence $a_1, a_2, a_3, \ldots\ldots a_{98}$ satisfies the relation $a_{n+1} = a_n + 1$ for $n = 1, 2, 3, \ldots\ldots 97$ and has the sum equal to 4949. Evaluate $\sum_{k=1}^{49} a_{2k}$.

श्रेणी $a_1, a_2, a_3, \ldots\ldots a_{98}$ सम्बन्ध $a_{n+1} = a_n + 1$ को $n = 1, 2, 3, \ldots\ldots 97$ के लिये संतुष्ट करती है एवं इसका योग 4949 है।

$\sum_{k=1}^{49} a_{2k}$ का मान निकालिए।

Ans. 2499

Sol. $S = a_1 + a_2 + a_3 + \ldots\ldots + a_{98} = 4949$
$a_{n+1} = a_n + 1 \Rightarrow d = a_{n+1} - a_n = 1$

$$S = \frac{98}{2}\left[2a_1 + (98-1)\cdot 1\right] = 4949$$

$\Rightarrow 2a + 97 = 101$

$2a = 4$

$\Rightarrow a = 2$

$$S' = \sum_{k=1}^{49} a_{2k} = \left[a_2 + a_4 + a_6 + ... + a_{98}\right]$$

$= [3 + 5 + 7 + ... + 99]$

$$= \frac{49}{2}\left[2\times 3 + (49-1)\times 2\right]$$

$= 2499$

8. The value of $x + y + z$ is 15 if a, x, y, z, b are in AP while the value of; $(1/x)+(1/y)+(1/z)$ is 5/3 if a, x, y, z, b are in HP. Find a and b.

$x + y + z$ का मान 15 है, यदि a, x, y, z, b AP में हों; जबकि $(1/x)+(1/y)+(1/z)$ का मान 5/3 है यदि a, x, y, z, b ह.श्रे. में हो तो a एवं b का मान निकालिये।

Ans. $a = 1, b = 9$ or $b = 1, a = 9$

Sol. $x + y + z = 15$ .....(i)

a, x, y, z, b are in AP

Suppose d is common difference $d = \frac{b-a}{4}$

$\therefore \quad x = a + \frac{b-a}{4} = \frac{b+3a}{4}, \qquad y = \frac{2b+2a}{4}$ and $z = \frac{3b+a}{4}$

on substituting the values of x, y and z in (i), we get

$\Rightarrow \quad \frac{6a+6b}{4} = 15$

$a + b = 10$ ..... (ii)

$\because \quad \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{5}{3}$ .....(iii)

and a, x, y, z, b are in H.P. $\therefore \quad \frac{1}{a}, \frac{1}{x}, \frac{1}{y}, \frac{1}{z}, \frac{1}{b}$ are in A.P.

$$\therefore \quad \frac{1}{x} = \frac{1}{a} + \frac{\left(\frac{1}{b} - \frac{1}{a}\right)}{4}, \quad \frac{1}{y} = \frac{1}{a} + \frac{2}{4}\left(\frac{1}{b} - \frac{1}{a}\right) \text{ and } \frac{1}{z} = \frac{1}{a} + \frac{3}{4}\left(\frac{1}{b} - \frac{1}{a}\right)$$

on substituting the value of $\frac{1}{x},\frac{1}{y},\frac{1}{z}$ in (iii), we get

$$\frac{3}{a}+\frac{6}{4}\left(\frac{1}{b}-\frac{1}{a}\right)=\frac{5}{3} \quad \Rightarrow \quad \frac{3}{2a}+\frac{3}{2b}=\frac{5}{3} \Rightarrow \frac{1}{a}+\frac{1}{b}=\frac{10}{9} \quad \text{..... (iv)}$$

By equations (ii) and (iv), we get

$a = 9, b = 1$ or $a = 1, b = 9$

Hindi. $x + y + z = 15$ .....(i)

a, x, y, z, b समान्तर श्रेढ़ी में है माना d सार्वअन्तर है

$$d = \frac{b-a}{4} \quad \therefore \quad x = a + \frac{b-a}{4} = \frac{b+3a}{4}, \quad y = \frac{2b+2a}{4} \quad \text{and} \quad z = \frac{3b+a}{4}$$

समीकरण (i) में x, y और z का मान रखने पर

$$\Rightarrow \quad \frac{6a+6b}{4} = 15$$

$$\Rightarrow \quad a + b = 10 \quad \text{..... (ii)} \quad \because \quad \frac{1}{x}+\frac{1}{y}+\frac{1}{z}=\frac{5}{3} \quad \text{.....(iii)}$$

तथा a, x, y, z, b हरात्मक श्रेढ़ी में है

$$\therefore \quad \frac{1}{a},\frac{1}{x},\frac{1}{y},\frac{1}{z},\frac{1}{b} \text{ समान्तर श्रेढ़ी में है}$$

$$\therefore \quad \frac{1}{x}=\frac{1}{a}+\frac{\left(\frac{1}{b}-\frac{1}{a}\right)}{4}, \quad \frac{1}{y}=\frac{1}{a}+\frac{2}{4}\left(\frac{1}{b}-\frac{1}{a}\right) \text{ तथा } \frac{1}{z}=\frac{1}{a}+\frac{3}{4}\left(\frac{1}{b}-\frac{1}{a}\right)$$

समीकरण (iii) में $\frac{1}{x},\frac{1}{y},\frac{1}{z}$ का मान रखने पर

$$\frac{3}{a}+\frac{6}{4}\left(\frac{1}{b}-\frac{1}{a}\right)=\frac{5}{3} \quad \Rightarrow \quad \frac{3}{2a}+\frac{3}{2b}=\frac{5}{3} \Rightarrow \frac{1}{a}+\frac{1}{b}=\frac{10}{9} \quad \text{..... (iv)}$$

समीकरण (ii) और (iv) से हमें प्राप्त होता है

$a = 9, b = 1$ या $a = 1, b = 9$

9. The values of xyz is 15/2 or 18/5 according as the series a, x, y, z, b is an AP or HP. Find the values of a and b assuming them to be positive integer.

xyz के मान 15/2 या 18/5 है जब a, x, y, z, b AP अथवा HP में क्रमशः है a, b को धनात्मक पूर्णांक मानते हुए इनका मान निकालिये।

Ans. $a = 1$; $b = 3$ or vice versa

Sol. If a, x, y, z, b are in A.P. then $xyz = \frac{15}{2}$

If a, x, y,z, b are in H.P. then $xyz = \frac{18}{5}$

$b = a + 4d \Rightarrow \left(\frac{b-a}{4}\right) = d$

$xyz = (a + d)(a + 2d)(a + 3d)$

$$= \left(a + \frac{b-a}{4}\right)\left(a + \frac{2b-2a}{4}\right)\left(a + \frac{3b-3a}{4}\right)$$

$$xyz = \frac{(3a+b)(2a+2a)(a+3b)}{64} = \frac{15}{2} \qquad ...(1)$$

If a, x, y, z, b are in H.P.

$\frac{1}{a}, \frac{1}{x}, \frac{1}{y}, \frac{1}{z}, \frac{1}{b}$ are in A.P.

$$\left(\frac{1}{b} - \frac{1}{a}\right) = 4d' \Rightarrow d' = \frac{1}{4}\left(\frac{1}{b} - \frac{1}{a}\right)$$

$$\frac{1}{xyz} = \left(\frac{1}{a} + d'\right)\left(\frac{1}{a} + 2d'\right)\left(\frac{1}{a} + 3d'\right)$$

$$= \left[\frac{1}{a} + \frac{1}{4}\left(\frac{1}{b} - \frac{1}{a}\right)\right]\left[\frac{1}{a} + \frac{2}{4}\left(\frac{1}{b} - \frac{1}{a}\right)\right]\left[\frac{1}{a} + \frac{3}{4}\left(\frac{1}{b} - \frac{1}{a}\right)\right]$$

$$\frac{5}{18} = \left[\frac{3b+a}{4ab}\right]\left[\frac{2a+2b}{4ab}\right]\left[\frac{b+3a}{4ab}\right] \qquad ...(2)$$

$$\frac{(1)}{(2)} : \frac{\left(\frac{15}{2}\right)}{\left(\frac{5}{18}\right)} = \frac{\left(\frac{(3a+b)(2a+2b)(a+3b)}{64}\right)}{\frac{(3a+b)(2a+2b)(a+3b)}{64(ab)^3}}$$

$$\frac{15}{2} \times \frac{18}{5} = (ab)^3 \Rightarrow ab = 3$$

Since, a, b are positive integers.

So, a = 1, b = 3 or a = 3, b = 1

10. If the roots of $10x^3 - cx^2 - 54x - 27 = 0$ are in harmonic progression, then find *c* and all the roots.

समीकरण $10x^3 - cx^2 - 54x - 27 = 0$ के मूल यदि ह. श्रे. में है तो c का मान निकालिए एवं सभी मूल निकालिए।

Ans. C = 9 ;(3, –3/2 , -3/5)11

Sol. $10x^3 - cx^2 - 54x - 27 = 0$

let α, β, γ are roots

$$\alpha+\beta+\gamma = \frac{c}{10} \qquad \alpha\beta+\beta\gamma+\gamma\alpha = \frac{-54}{10} \qquad \alpha\beta\gamma = \frac{27}{10}$$

α, β, γ are in H.P.

$$\Rightarrow B = \frac{2\alpha\gamma}{\alpha+\gamma} \Rightarrow \alpha+\gamma = \frac{2\alpha\gamma}{B}$$

$$\alpha\beta+\beta\gamma+\gamma\alpha = \frac{-54}{10}$$

$$B(\alpha+\gamma)+\alpha\gamma = \frac{-54}{10}$$

$$B\left(\frac{2\gamma\alpha}{B}\right)+\gamma\alpha = \frac{-54}{10}$$

$$3\gamma\alpha = \frac{-54}{10} \Rightarrow \gamma\alpha = \frac{-18}{10} \Rightarrow B = \frac{-3}{2}$$

$$\alpha\beta+\beta\gamma+\gamma\alpha = \frac{-54}{10}$$

$$B(\alpha+\gamma)-\frac{18}{10} = \frac{-54}{10}$$

$$B(\alpha+\gamma) = \frac{-36}{10}$$

$$(\alpha+\gamma) = \frac{12}{5}$$

$x^2 - (\alpha+\gamma)x + \alpha\gamma = 0$

$10x^2 - 24x - 18 = 0 \Rightarrow (10x+6)(x-3) = 0$

$$x = \frac{-6}{10},\ x = 3$$

$$x = \frac{-3}{5},\ x = 3$$

let $\alpha = \frac{-3}{5}, \alpha = 3$

$$\alpha+\beta+\gamma = \frac{+c}{10}$$

$$\frac{-3}{5}+\left(\frac{-3}{2}\right)+3 = \frac{c}{10}$$

$$\frac{-6-15+30}{10} = \frac{c}{10}$$

$c = 9$

11. In a GP the ratio of the sum of the first eleven terms to the sum of the last eleven terms is 1/8 and the ratio of the sum of all the terms without the first nine to the sum of all the terms without the last nine is 2. Find the number of terms in the GP.

एक गु. श्रे. में प्रथम ग्यारह पदों के योग एवं अन्तिम ग्यारह पदों के योग का अनुपात 1/8 है तथा प्रथम नौ पदों को छोड़कर अन्य सभी पदों के योग एवं अन्तिम नौ पदों को छोड़कर अन्य सभी पदों के योग का अनुपात 2 है। गु. श्रे. में पदों की संख्या बताइए।

Ans. $n = 38$

Sol. $$\frac{a_1 + a_2 + ... + a_{11}}{a_{n-10} + a_{n-9} + ... + a_n} = \frac{1}{8}$$

$$\frac{a + ar + ... + ar^{10}}{ar^{n-11} + ar^{n-10} + ... + ar^{n-1}} = \frac{1}{8}$$

$$\frac{a\left(\frac{r^{11}-1}{r-1}\right)}{ar^{n-11}\left(\frac{r^{11}-1}{r-1}\right)} = \frac{1}{8}$$

$r^{n-11} = 2^3$ ...(1)

$$\frac{S_n - (a_1 + a_2 + ... + a_9)}{S_n - (a_{n-8} + a_{n-7} + ... + a_n)} = \frac{2}{1}$$

$$\frac{a\left(\frac{r^n-1}{r-1}\right) - a\left(\frac{r^9-1}{r-1}\right)}{a\left(\frac{r^n-1}{r-1}\right) - \left(\left(a\left(r^{n-9}\right)\right)\left(\frac{r^9-1}{r-1}\right)\right)} = \frac{2}{1}$$

$$\frac{r^n - r^9}{-1 + r^{n-9}} = \frac{2}{1}$$

$$r^9\left(\frac{r^{n-9}-1}{r^{n-9}-1}\right) = 2$$

$r^9 = 2$ ...(2)

By (1), (2)

$r^{n-11} = 2^3 = (r^9)^3$

$r^{n-11} = r^{27} \Rightarrow n = 27 + 11 = 38$

12. Let a_n be a sequence such that $a_1 = 3,\ a_{n+1} = 3a_n + 1$ $(n = 1, 2, 3, ...)$. If the value of $\sum_{n=1}^{\infty} \frac{a_n}{5^n} = \frac{p}{q}$. (where p and q are in their lowest form), then find the value of $(p + q)$.

यदि a_n श्रृंखला (sequence) इस प्रकार से है कि $a_1 = 3,\ a_{n+1} = 3a_n + 1$ $(n = 1, 2, 3, ...)$ यदि $\sum_{n=1}^{\infty} \frac{a_n}{5^n} = \frac{p}{q}$ तो $p + q$ का मान बताओं जहाँ p व q का महत्तम समापवर्तक 1 है।

Ans. 21

Sol. $a_1 = 3,\ a_{n+1} = 3a_{n+1}$

$a_1, a_2, a_3, a_4, ...$

$3, 3^2 + 1, 3^3 + 4, 3^4 + 13, ...$

$$\sum_{n=1}^{\infty} \frac{a_n}{5^n} = \frac{3}{5} + \frac{3^2+1}{5^2} + \frac{3^3+4}{5^3} + \frac{3^4+13}{5^4} + \frac{3^5+40}{5^5} + ...$$

$$= \underbrace{\left[\frac{3}{5} + \left(\frac{3}{5}\right)^2 + \left(\frac{3}{5}\right)^3 + \left(\frac{3}{5}\right)^4 + \left(\frac{3}{5}\right)^5 + ...\right]}_{S_1} \underbrace{\left[\frac{1}{5^2} + \frac{4}{5^3} + \frac{13}{5^4}\right]}_{S_2}$$

$$S_1 = \frac{\frac{3}{5}}{1 - \frac{3}{5}} = \frac{3}{2}$$

$$S_2 = \frac{1}{5^2} + \frac{4}{5^3} + \frac{13}{5^4} + \frac{40}{5^5} + ... \quad ...(1)$$

$$\frac{S_2}{5} = 0 + \frac{1}{5^3} + \frac{4}{5^4} + \frac{13}{5^5} + ... \quad ...(2)$$

$(1) - (2)$:

$$\frac{4S_2}{5} = \left(\frac{1}{5^2}\right) + \frac{3}{5^3} + \frac{9}{5^4} + \frac{27}{5^5} + ...$$

$$\frac{4S_2}{5} = \frac{1}{25} + \frac{1}{9}\left[\left(\frac{3}{5}\right)^3 + \left(\frac{3}{5}\right)^4 + \left(\frac{3}{5}\right)^5 + ...\right]$$

$$= \frac{1}{25} + \frac{1}{9}\left[\frac{\left(\frac{3}{5}\right)^3}{1 - \frac{3}{5}}\right]$$

$$\frac{4S_2}{5}=\frac{1}{25}+\frac{1}{9}\left[\frac{27\times5}{125\times2}\right]=\frac{1}{10}$$

$$S_2=\frac{1}{8}$$

$$S=\frac{1}{8}+\frac{3}{2}=\frac{13}{8}=\frac{p}{q}\Rightarrow p+q=21$$

13. Solve the following equations for x and y

$$\log_2 x+\log_4 x+\log_{16} x+\ldots\ldots\ldots\ldots=y$$

$$\frac{5+9+13+\ldots\ldots\ldots+(4y+1)}{1+3+5+\ldots\ldots\ldots+(2y-1)}=4\log_4 x$$

x तथा y के लिये निम्नलिखित समीकरणों को हल कीजिये।

$$\log_2 x+\log_4 x+\log_{16} x+\ldots\ldots\ldots\ldots=y$$

$$\frac{5+9+13+\ldots\ldots\ldots+(4y+1)}{1+3+5+\ldots\ldots\ldots+(2y-1)}=4\log_4 x$$

Ans. $x=2\sqrt{2}$ and $y=3$

Sol. $\log_2 x+\log_4 x+\log_{16} x+\ldots\ldots=y$

$$\log_2 x+1\log_2 x+\frac{1}{4}\log_2 x+\ldots\ldots=y$$

$$\log_2 x+\log_2 x^{\frac{1}{2}}+\log_2 x^{\frac{1}{4}}+\ldots\ldots=y$$

$$\log_2 x^{\left(1+\frac{1}{2}+\frac{1}{4}+\ldots\ldots\right)}=y$$

$$\left(\log_2 x\right)\left(\frac{1}{1-\frac{1}{2}}\right)=y$$

$2\log_2 x=y$ _____(1)

$$\frac{5+9+13+\ldots\ldots+(4y+1)}{1+3+5+\ldots\ldots+(2y-1)}=4\log_4 x=2\log_2 x=y \quad \text{(By (1))}$$

$$\frac{5+9+13+\ldots\ldots+(4y+1)}{1+3+5+\ldots\ldots+(2y-1)}=y$$ _____(2)

$S = 5 + 9 + 13 + \ldots\ldots\ldots + (4y + 1) = \frac{y}{2}\left[2\times5+\left(y-1\right)\times4\right] = y\,(2y + 3)$

$S' = 1 + 3 + 5 + \ldots\ldots\ldots + (2y - 1) = \frac{y}{2}\left[2\times1+\left(y-1\right)\times2\right] = y^2$

By equation (2)

$$\frac{y\left(3+2y\right)}{y^2} = y$$

$y^2 - 2y - 3 = 0\ (y \neq 0)$

$$y = \frac{2 \pm \sqrt{4+12}}{2}$$

$$y = \frac{2 \pm 4}{2}$$

$y = 3, -1$ (reject)

$\Rightarrow y = 3$

$2\log_2 x = y = 3$

$$x = \left(2\right)^{\frac{3}{2}}$$

$$x = 2\sqrt{2}$$

14. Find the sum in the nth group of sequence :

निम्न अनुक्रमों के n वें समूह का योगफल ज्ञात कीजिए :

(i) 1, (2, 3); (4, 5, 6, 7); (8, 9,....., 15); ..... (ii) (1), (2, 3, 4), (5, 6, 7, 8, 9),.....

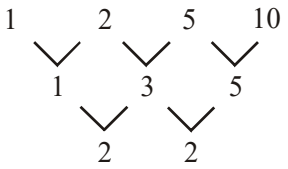
Ans. (i) $2^{n-2}\,(2^n + 2^{n-1} - 1)$ (ii) $(n - 1)^3 + n^3$

Sol. (i) 1, (2, 3), (4, 5, 6, 7), (8, 9, ..., 15) ....

Number of terms in n^{th} group $= 2^{n-1}$; 1st term in n^{th} group $= 2^{n-1}$

So Sum of terms in n^{th} group $= \frac{2^{n-1}}{2}[2.2^{n-1} + (2^{n-1} - 1)\ 1] = 2^{n-2}\,[2^n + 2^{n-1} - 1]$

(ii) (1) , (2, 3, 4), (5, 6, 7, 8, 9) ; 1st term in n^{th} group let T_n



$T_n = a + bn + cn^2 \Rightarrow T_1 = a + b + c = 1$ .... (i)

$T_2 = a + 2b + 4c = 2$ .... (ii)

and $T_3 = a + 3b + 9c = 5$ .... (iii)

$a = 2, b = -2, c = 1$

On solving (i), (ii) and (iii), we get

So Ist term is $T_n = (2 - 2n + n^2)$. Number of terms in n^{th} group $= (2n - 1)$

$\therefore$ sum of terms in n^{th} group $= \frac{2n-1}{2}\,[2\,(2 - 2n + n^2) + 2n - 2]$

$= (2n - 1)\,[n^2 - n + 1] = 2n^3 - 2n^2 + 2n - n^2 + n - 1 = n^3 + (n - 1)^3$

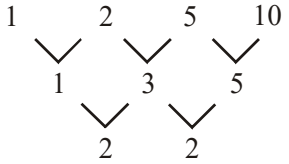
Hindi (i) 1, (2, 3), (4, 5, 6, 7), (8, 9, ..., 15) ....

n वें समूह में पदों की संख्या $= 2^{n-1}$

n वें समूह का प्रथम पद $= 2^{n-1}$

इसलिए n वें समूह के पदों का योग $= \frac{2^{n-1}}{2}\,[2.2^{n-1} + (2^{n-1} - 1)\,1] = 2^{n-2}\,[2^n + 2^{n-1} - 1]$

(ii) (1) , (2, 3, 4), (5, 6, 7, 8, 9) माना n वें समूह का प्रथम पद T_n है



$T_n = a + bn + cn^2 \Rightarrow T_1 = a + b + c = 1$ .... (i)

$T_2 = a + 2b + 4c = 2$ .... (ii)

और $T_3 = a + 3b + 9c = 5$ .... (iii)

$a = 2, b = -2, c = 1$

(i), (ii) और (iii) को हल करने पर

इसलिए प्रथम पद $T_n = (2 - 2n + n^2)$ $\Rightarrow$ n वें समूह में पदों की संख्या $= (2n - 1)$

$\therefore$ n वें समूह के पदों का योग $= \frac{2n-1}{2}\,[2\,(2 - 2n + n^2) + 2n - 2] = (2n - 1)\,[n^2 - n + 1]$

$= 2n^3 - 2n^2 + 2n - n^2 + n - 1 = n^3 + (n - 1)^3$

15. If a, b and c are in arithmetic progression and a^2, b^2 & c^2 are in harmonic progression, then prove that either $a = b = c$ or a, b and $-\frac{c}{2}$ are in geometric progression.

यदि a, b, c स.श्रे. में हैं और a^2, b^2, c^2 ह.श्रे. में हो, तो सिद्ध कीजिए कि या तो $a = b = c$ या a, b, $-\frac{c}{2}$ गु.श्रे. में हैं।

Ans.

Sol. a, b, c are in A.P. $\Rightarrow 2b = a + c$

a^2, b^2, c^2 are in H.P. $\Rightarrow b^2 = \frac{2a^2c^2}{a^2 + c^2}$

$$\left(\frac{a+c}{2}\right)^2 = \frac{2a^2c^2}{a^2+c^2}$$

$(a^2 + c^2)\,[(a^2 + c^2) + 2ac] = 8a^2c^2$

$(a^2 + c^2)^2 + 2a^3c + 2a^2c^3 = 8a^2c^2$

$a^4 + c^4 + 2a^3c + 2a^2c^3 - 6a^2c^2 = 0$

$[(a^2)^2 + (c^2)^2 - 2a^2c^2] + 2ac\,[a^2 + c^2 - 2ac] = 0$

$(a^2 - c^2)^2 + 2ac\,[a - c]^2 = 0$

$(a - c)^2\,[(a + c)^2 + 2ac] = 0$

$(a - c)^2\,[a^2 + 4ac + c^2] = 0$

$a - c = 0$ or $a^2 + 4ac + c^2 = 0$

$a = c$ or $(a^2 + 2ac + c^2) + 2ac = 0$

$(a + c)^2 + 2ac = 0$

$(2b)^2 = -2ac$

$$b^2 = (a)\left(\frac{-c}{2}\right)$$

a, b, $-\frac{c}{2}$ are in G.P.

16. Let 'X' denotes the value of the product $(1 + a + a^2 + a^3 + ... \infty)(1 + b + b^2 + b^3 + ... \infty)$ where 'a' and 'b' are the roots of the quadratic equation $11x^2 - 4x - 2 = 0$ and 'Y' denotes the numerical value of the infinite series

$$(\log_b 2)^0 \left(\log_b 5^{4^0}\right) + (\log_b 2)^1 \left(\log_b 5^{4^1}\right) + (\log_b 2)^2 \left(\log_b 5^{4^2}\right) + (\log_b 2)^3 \left(\log_b 5^{4^3}\right) + ...\infty$$

where b = 2000. Find (XY).

दिया है कि 'X' श्रेणियों

$(1 + a + a^2 + a^3 + ... \infty)(1 + b + b^2 + b^3 + ... \infty)$

के गुणन को व्यक्त करता है जहाँ 'a' और 'b' द्विघात समी. $11x^2 - 4x - 2 = 0$ के मूल है तथा 'Y' श्रेणी

$$(\log_b 2)^0 \left(\log_b 5^{4^0}\right) + (\log_b 2)^1 \left(\log_b 5^{4^1}\right) + (\log_b 2)^2 \left(\log_b 5^{4^2}\right) + (\log_b 2)^3 \left(\log_b 5^{4^3}\right) + ...\infty$$

के आंकिक मान को व्यक्त करता है जहाँ b = 2000 तब XY का मान ज्ञात कीजिए।

Ans. $\frac{11}{15}$

Sol. $X = (1 + a + a^2 + ...)\,(1 + b + b^2 + ...) = \left(\frac{1}{1-a}\right)\left(\frac{1}{1-b}\right)$

$11x^2 - 4x - 2 = 0$

Now we form a quadratic equation whose roots are $\frac{1}{1-a}, \frac{1}{1-b}$

let x = a

$$y = \frac{1}{1-a} = \frac{1}{1-x}$$

$$x = \frac{y-1}{y}$$

$$11\left(\frac{y-1}{y}\right)^2 - 4\left(\frac{y-1}{y}\right) - 2 = 0$$

$5y^2 - 18y + 11 = 0$

$$X = \left(\frac{1}{1-a}\right)\left(\frac{1}{1-b}\right) = \frac{11}{5}$$

$$Y = (\log_b 2)^0 \left(\log_b 5^{4^0}\right) + (\log_b 2)^1 \left(\log_b 5^{4^1}\right) + (\log_b 2)^2 \left(\log_b 5^{4^2}\right) + ...$$

$$Y = \log_b 5 + 4(\log_b 2)^1 (\log_b 5) + 4^2 (\log_b 2)(\log_b 5) + ...$$

$$Y = \frac{\log_b 5}{1 - 4\log_b 2} = \frac{\log_2 b \times \log_b 5}{\log_2 b - 4} = \frac{\log_2 5}{\log_2 \frac{b}{16}}$$

At b = 2000

$$Y = \frac{\log_2 5}{\log_2 \frac{2000}{16}} = \frac{\log_2 5}{3\log_2 5} = \frac{1}{3}$$

$$XY = \left(\frac{11}{5}\right)\left(\frac{1}{3}\right) = \frac{11}{15}$$

17. The sum of the squares of three distinct real numbers, which are in GP is S^2. If their sum is αS, show that $\alpha^2 \in (1/3, 1) \cup (1, 3)$.

तीन भिन्न वास्तविक संख्याओं, जो गु.श्रे. में है, का योग αS तथा उनके वर्गों का योग S^2 है तो सिद्ध कीजिये $\alpha^2 \in (1/3, 1) \cup (1, 3)$.

Ans.

Sol. $a + b + c = \alpha S$, $a^2 + b^2 + c^2 = S^2$

$a^2 + b^2 + c^2 + 2(ab + bc + ca) = \alpha^2 (a^2 + b^2 + c^2)$

$2(ab + bc + ca) = (\alpha^2 - 1)(a^2 + b^2 + c^2)$

a, b, c are in G.P. $\Rightarrow$ a, ar, ar^2

$2a^2 (r + r^2 + r^3) = a^2 (\alpha^2 - 1)(1 + r^2 + r^4)$

$2r(1 + r + r^2) = (\alpha^2 - 1)(1 + r + r^2)(1 - r + r^2)$

$$\frac{2}{1 - r + r^2} + 1 = \alpha^2$$

let $\alpha^2 = y$

$$y = \frac{1+r+r^2}{1-r+r^2}$$

$(y-1)\,r^2 + (-y-1)r + (y-1) = 0$

Case–1 $y \neq 1$ $D \geq 0$

$(1+y)^2 - 4(y-1)^2 \geq 0$

$y^2 + 3y - 1 \geq 0$

$$y \in \left(\frac{1}{3}, 3\right)$$

Case–2 $y = 1 \Rightarrow \alpha^2 = 1$

$1 + r + r^2 = 1 - r + r^2$

$2r = 0$

$\Rightarrow r = 0$

But $r \neq 0$ ($\because$ a, b, c are in G.P.)

Hence $y = \alpha^2 \in \left[\frac{1}{3}, 1\right) \cup (1, 3]$

18. Find the sum to 'n' terms of the series

$$\frac{3}{1.2}\cdot\frac{1}{2} + \frac{4}{2.3}\cdot\frac{1}{2^2} + \frac{5}{3.4}\cdot\frac{1}{2^3} + \frac{6}{4.5}\cdot\frac{1}{2^4} + \ldots\ldots\ldots$$

श्रेणी $\frac{3}{1.2}\cdot\frac{1}{2} + \frac{4}{2.3}\cdot\frac{1}{2^2} + \frac{5}{3.4}\cdot\frac{1}{2^3} + \frac{6}{4.5}\cdot\frac{1}{2^4} + \ldots\ldots\ldots$ के n पदों का योग ज्ञात कीजिये।

Ans. $\left(1 - \frac{1}{(n+1)\,2^n}\right)$

Sol. $S = \frac{3}{1\cdot2}\cdot\frac{1}{2} + \frac{4}{2\cdot3}\cdot\frac{1}{2^2} + \frac{5}{3\cdot4}\cdot\frac{1}{2^3} + \frac{6}{4\cdot5}\cdot\frac{1}{2^4} + \ldots\ldots$

$$T_n = \frac{(n+2)}{n(n+1)}\frac{1}{2^n}$$

$$T_n = \left[\frac{(2n+2)-n}{n(n+1)}\right]\frac{1}{2^n}$$

$$= \left[\left(\frac{2}{n}\right) - \left(\frac{1}{n+1}\right)\right]\frac{1}{2^n}$$

$$T_n = \frac{1}{n\cdot 2^{n-1}} - \frac{1}{(n+1)2^n}$$

$$S_n = \sum_{r=1}^{n}\left[\frac{1}{r\cdot 2^{r-1}} - \frac{1}{(r+1)2^r}\right]$$

$$S_n = \left(\frac{1}{1\cdot 2^{1-1}} - \frac{1}{2\cdot 2^1}\right) + \left(\frac{1}{2\cdot 2} - \frac{1}{3\cdot 2^2}\right) + ... + \left(\frac{1}{n\cdot 2^{n-1}} - \frac{1}{(n+1)2^n}\right)$$

$$S_n = 1 - \frac{1}{(n+1)2^n}$$

19. Find the sum of infinite terms of the series : $\frac{3}{2.4} + \frac{5}{2.4.6} + \frac{7}{2.4.6.8} + \frac{9}{2.4.6.8.10} + ...$

श्रेणी $\frac{3}{2.4} + \frac{5}{2.4.6} + \frac{7}{2.4.6.8} + \frac{9}{2.4.6.8.10} + ...$ के अनन्त पदों का योग ज्ञात कीजिये।

Ans. 1/2

Sol. $S = \frac{3}{2\cdot 4} + \frac{5}{2\cdot 46} + \frac{7}{2\cdot 4\cdot 6\cdot 8} + \frac{9}{2\cdot 4\cdot 6\cdot 8\cdot 10} + ...$

$$T_r = \frac{2r+1}{2\cdot 4.....(2r+2)} = \frac{(2r+2)-1}{2\cdot 4.....(2r+2)}$$

$$T_r = \frac{1}{2\cdot 4\cdot .....2r} - \frac{1}{2\cdot 4\cdot .....(2r+2)}$$

$$S_\infty = \left(\frac{1}{2} - \frac{1}{2\cdot 4}\right) + \left(\frac{1}{2\cdot 4} - \frac{1}{2\cdot 4\cdot 6}\right) + \left(\frac{1}{2\cdot 4\cdot 6} - \frac{1}{2\cdot 4\cdot 6\cdot 8}\right) + ...$$

$$= \frac{1}{2}$$

20. Let S_k, k = 1, 2, ..., 100, denote the sum of the infinite geometric series whose first term is $\frac{k-1}{k!}$ and the common ratio is $\frac{1}{k}$. Then the value of $\frac{100^2}{100!} + \sum_{k=1}^{100}\left|\left(k^2 - 3k + 1\right)S_k\right|$, is.

यदि S_k किसी अनन्त गुणोत्तर श्रेणी का योग है जहाँ k = 1, 2, ..., 100, जिसका प्रथम पद $\frac{k-1}{k!}$ तथा सर्वानुपात $\frac{1}{k}$ है तो

$\frac{100^2}{100!}+\sum_{k=1}^{100}\left|\left(k^2-3k+1\right)S_k\right|$ का मान बताओं।

Ans. 3

Sol. $S_k=\frac{\left(\frac{k-1}{k!}\right)}{1-\frac{1}{k}}=\frac{\frac{k-1}{k!}}{\frac{(k-1)}{k}}=\frac{1}{(k-1)!}$

$$A=\frac{(100)^2}{100!}+\sum_{k=1}^{100}\left|\left(k^2-3k+1\right)S_k\right| \qquad ...(1)$$

$$=\frac{100^2}{100!}+\sum_{k=1}^{100}\left|\left(k^2-3k+1\right)\frac{1}{(k-1)!}\right|$$

$$=\frac{1}{99!}+\frac{1}{98!}+\sum_{k=1}^{100}\left|\frac{(k-2)(k-1)-1}{(k-1)!}\right|$$

$$=\frac{1}{99!}+\frac{1}{98!}+\sum_{k=3}^{100}\left(\frac{1}{(k-3)!}-\frac{1}{(k-1)!}\right)+(1-3+1)\times\frac{1}{(1-1!)}+(4-6+2)\times\frac{1}{(2-1)!}$$

$$=\frac{1}{99!}+\frac{1}{98!}+1+1-\frac{1}{99!}-\frac{1}{98!}+1+0$$

$$= 1 + 1 + 1 = 3$$

21. If a, b, c are positive real number such that $\frac{\log a}{b-c}=\frac{\log b}{c-a}=\frac{\log c}{a-b}$ then prove that $a^{b+c}+b^{c+a}+c^{a+b}\geq 3$.

यदि a, b, c धनात्मक वास्तविक संख्याऐं है इस प्रकार कि $\frac{\log a}{b-c}=\frac{\log b}{c-a}=\frac{\log c}{a-b}$ तो सिद्ध कीजिये कि

$a^{b+c}+b^{c+a}+c^{a+b}\geq 3$.

Ans.

Sol. $\frac{\log a}{b-c}-\frac{\log b}{c-a}=\frac{\log c}{a-b}=k$

$(\log a)(c-a)=(b-c)(\log b)$

$[a\log a+b\log b]=c[\log a+\log b]$

$\log a^a b^b = \log (ab)^c$

$a^a b^b = a^c b^c$

Similarly

$b^a c^a = b^b c^c$

and

$a^a c^a = a^b c^b$

$$\frac{a^{b+c} + b^{c+a} + c^{a+b}}{3} \geq \left(a^{b+c} \cdot b^{c+a} \cdot c^{a+b}\right)^{\frac{1}{2}}$$

$$= \left[\left(a^b c^b\right)\left(a^c b^c\right)\left(b^a c^a\right)\right]^{\frac{1}{2}}$$

$$= \left[\left(a^a c^a\right)\left(a^a b^b\right)\left(b^b c^c\right)\right]^{\frac{1}{2}}$$

$= a^a \cdot b^b c^c$

$\Rightarrow a^{b+c} + b^{c+a} + c^{a+b} \geq 3(a^a b^b c^c)$ _____(1)

$\log a = k\,(b - c)$

$a = e^{k(b-c)}$, $b = e^{k(c-a)}$, $c = e^{k(a-b)}$

$a^a b^b c^c = e^{k[ab - ac + bc - ab + ac - cb]} = 1$

Put value in (1)

$a^{b+c} + b^{c+a} + c^{a+b} \geq 3$

22. If positive square root of, $a^{\frac{1}{a}} \cdot (2a)^{\frac{1}{2a}} \cdot (4a)^{\frac{1}{4a}} \cdot (8a)^{\frac{1}{8a}} \ldots\ldots\ldots\ldots\infty$ is $\frac{8}{27}$, find the value of 'a'.

यदि $a^{\frac{1}{a}} \cdot (2a)^{\frac{1}{2a}} \cdot (4a)^{\frac{1}{4a}} \cdot (8a)^{\frac{1}{8a}} \ldots\ldots\ldots\ldots\infty$ का धनात्मक वर्ग मूल $\frac{8}{27}$ है तो 'a' का मान ज्ञात कीजिये।

Ans. $a = \frac{1}{3}$

Sol. $A = \left(a^{\frac{1}{a}}\right)(2a)^{\frac{1}{2a}}(4a)^{\frac{1}{4a}} \cdot (8a)^{\frac{1}{8a}} \ldots\ldots\ldots$

$$= a^{\left(\frac{1}{a} + \frac{1}{2a} + \frac{1}{4a} + \frac{1}{8a} + \ldots\ldots\right) \cdot 2\left(\frac{1}{2a} + \frac{2}{4a} + \frac{3}{8a} + \ldots\ldots\ldots\right)}$$

$$= a^{\frac{1}{a}\left[\frac{1}{1-\frac{1}{2}}\right] \cdot 2^{\frac{1}{a}\left[\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \ldots\ldots\right]}}$$ _____(1)

$$S = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \ldots\ldots\ldots$$

$$\frac{S}{2}=0+\frac{1}{2^2}+\frac{2}{2^3}+.....$$

$$\frac{S}{2}=\frac{1}{2}+\left(\frac{1}{2^2}+\frac{1}{2^3}+.....\right)$$

$$S=1+\frac{1}{2}\left[1+\frac{1}{2}+.....\right]$$

$$=1+\frac{1}{2}\left[\frac{1}{1-\frac{1}{2}}\right]$$

S = 1 + 1 = 2

By (1)

$$A=a^{\frac{2}{a}}\cdot 2^{\frac{1}{a}\times 2}$$

$$A=a^{\frac{2}{a}}\cdot 2^{\frac{2}{a}}=(2a)^{\frac{2}{a}}$$

$$\sqrt{A}=(2a)^{\frac{1}{a}}=\left(\frac{8}{27}\right)$$

$$a=\frac{1}{3}$$

23. Find the sum of the series , $\frac{n}{1\,.\,2\,.\,3}+\frac{n-1}{2\,.\,3\,.\,4}+.....+\frac{1}{n\,(n+1)\,(n+2)}$.

श्रेणी $\frac{n}{1\,.\,2\,.\,3}+\frac{n-1}{2\,.\,3\,.\,4}+.....+\frac{1}{n\,(n+1)\,(n+2)}$ का मान ज्ञात कीजिये।

Ans. $S=\frac{n\,(n+1)}{4\,(n+2)}$

Sol. $\frac{n}{1.2.3}+\frac{n-1}{2.3.4}+.....+\frac{1}{n(n+1)(n+2)}$; $S_n=\sum_{r=1}^{n}\frac{n+1}{r(r+1)(r+2)}-\sum_{r=1}^{n}\frac{1}{(r+1)(r+2)}$

Now अब $\sum_{r=1}^{n}\frac{1}{r(r+1)(r+2)}=\sum_{r=1}^{n}\frac{1}{2(r+1)}\left[\frac{1}{r}-\frac{1}{r+2}\right]=\frac{1}{2}\left[\sum_{r=1}^{n}\frac{1}{r(r+1)}-\sum_{r=1}^{n}\frac{1}{(r+1)(r+2)}\right]$

$$= \frac{1}{2}\left[\sum_{r=1}^{n}\left(\frac{1}{r}-\frac{1}{r+1}\right)-\sum_{r=1}^{n}\left(\frac{1}{r+1}-\frac{1}{r+2}\right)\right]=\frac{1}{2}\left[\left(1-\frac{1}{n+1}\right)-\left(\frac{1}{2}-\frac{1}{n+2}\right)\right]$$

$$= \frac{1}{2}\left[\frac{n}{n+1}-\frac{n}{2(n+2)}\right]=\frac{n}{2}\left[\frac{1}{n+1}-\frac{1}{2(n+2)}\right]$$

Now अब $S_n=(n+1)\frac{n}{2}\left[\frac{1}{n+1}-\frac{1}{2(n+2)}\right]-\left(\frac{1}{2}-\frac{1}{n+2}\right)$

$$=\frac{n}{2}-\frac{n(n+1)}{4(n+2)}-\frac{n}{2(n+2)}=\frac{2n^2+4n-n^2-n-2n}{4(n+2)}=\frac{n^2+n}{4(n+2)}=\frac{n(n+1)}{4(n+2)}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 24 to 25

The first four terms of a sequence are given by $T_1 = 0, T_2 = 1, T_3 = 1, T_4 = 2$.
The general term is given by $T_n = A\alpha^{n-1} + B\beta^{n-1}$ where A, B, α, β are independent of n and A is positive.

श्रेणी के प्रथम चार पद $T_1 = 0, T_2 = 1, T_3 = 1, T_4 = 2$ द्वारा दिये गये है।
श्रेणी का सामान्य पद $T_n = A\alpha^{n-1} + B\beta^{n-1}$ द्वारा दिया जाता है, जहां A, B, α तथा β, n से स्वतंत्र है तथा A धनात्मक है।

24. The value of $(\alpha^2 + \beta^2 + \alpha\beta)$ is equal to :

$(\alpha^2 + \beta^2 + \alpha\beta)$ का मान होगा :

(A) 1 (B) 2 (C) 5 (D) 4

Ans. B

Sol. $T_1 = 0, T_2 = 1, T_3 = 1, T_4 = 2$
$T_n = A\alpha^{n-1} + B\beta^{n-1}$
$T_1 = 0 = A + B \Rightarrow A = -B$
$T_2 = A\alpha + B\beta = 1 \Rightarrow A(\alpha) + (-A)\beta = 1 \Rightarrow A(\alpha - \beta) = 1$ ...(1)
$T_3 = A\alpha^2 + B\beta^2 = 1 \Rightarrow A(\alpha^2 - \beta^2) = 1$ ...(2)
$T_4 = A\alpha^3 + B\beta^3 = 2 \Rightarrow A(\alpha^3 - \beta^3) = 2$ ...(3)

$$\frac{(1)}{(2)}:\frac{A(\alpha-\beta)}{A(\alpha^2-\beta^2)}=\frac{1}{1}\Rightarrow \alpha+\beta=1$$

$$\frac{(1)}{(3)}:\frac{A(\alpha-\beta)}{A(\alpha^3-\beta^3)}=\frac{1}{2}$$

$$\frac{1}{\alpha^2+\beta^2+\alpha\beta}=\frac{1}{2}$$

$\alpha^2 + \beta^2 + \alpha\beta = 2$

$$\frac{(2)}{(3)}:\frac{A(\alpha-\beta)(\alpha+\beta)}{A(\alpha-\beta)(\alpha^2+\beta^2+\alpha\beta)}=\frac{1}{2}\Rightarrow\alpha+\beta=1$$

$(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta = 1$
$\alpha\beta = 1 - (\alpha^2 + \beta^2 + \alpha\beta)$
$\alpha\beta = -1$
$(\alpha - \beta)2 = \alpha^2 + \beta^2 - 2\alpha\beta$
$= (\alpha + \beta)^2 - 4\alpha\beta$
$= 1 + 4$
$(\alpha - \beta)^2 = 5$

$$\left(\frac{1}{A}\right)^2=5\Rightarrow A^2=\frac{1}{5}$$

25. The value of $5(A^2 + B^2)$ is equal to :

$5(A^2 + B^2)$ का मान होगा :

(A) 2 (B) 4 (C) 6 (D) 8

Ans. A

Sol. $5\left(A^2+B^2\right)=5\times2\times\frac{1}{5}=2$

Paragraph for Question Nos. 26 to 28

Let x, y, z are positive reals and $x + y + z = 60$ and $x > 3$.

माना x, y, z धनात्मक वास्तविक है तथा $x + y + z = 60$ तथा $x > 3$ है।

26. Maximum value of $(x - 3)(y + 1)(z + 5)$ is :

$(x - 3)(y + 1)(z + 5)$ का अधिकतम मान होगा :

(A) (17)(21)(25) (B) (20)(21)(23) (C) (21)(21)(21) (D) (23)(19)(15)

Ans. C

Sol. $x + y + z = 60, \quad x > 3$

$$\frac{(x-3)+(y+1)+(z+5)}{3}\geq\left[(x-3)(y+1)(z+5)\right]^{\frac{1}{3}}$$

$$\left(\frac{60-3+1+5}{3}\right)^3\geq(x-3)(y+1)(z+5)$$

$(21)^3 \geq (x - 3)(y + 1)(z + 5)$

27. Maximum value of $(x-3)(2y+1)(3z+5)$ is :

(A) $\frac{(355)^3}{3^3.6^2}$ (B) $\frac{(355)^3}{3^3.6^3}$ (C) $\frac{(355)^3}{3^2.6^3}$ (D) None of these

$(x-3)(2y+1)(3z+5)$ का अधिकतम मान होगा :

(A) $\frac{(355)^3}{3^3.6^2}$ (B) $\frac{(355)^3}{3^3.6^3}$ (C) $\frac{(355)^3}{3^2.6^3}$ (D) None of these

Ans. A

Sol. $$\frac{(x-3)+\left(y+\frac{1}{2}\right)+\left(z+\frac{5}{3}\right)}{3} \geq \left[(x-3)\left(y+\frac{1}{2}\right)\left(z+\frac{5}{3}\right)\right]^{\frac{1}{3}}$$

$$\frac{\left(60-3+\frac{1}{2}+\frac{5}{3}\right)^3}{3} \geq \frac{(x-3)(2y+1)(3z+5)}{6}$$

$$\frac{(355)^3}{3^3 6^2} \geq (x-3)(2y+1)(3z+5)$$

28. Maximum value of xyz is :

xyz का अधिकतम मान होगा :

(A) 8×10^3 (B) 27×10^3 (C) 64×10^3 (D) 125×10^3

Ans. A

Sol. $$\frac{x+y+z}{3} \geq (xyz)^{\frac{1}{3}}$$

$$\frac{60}{3} \geq (xyz)^{\frac{1}{3}}$$

$$(20)^3 \geq xyz$$

$$8 \times 10^3 \geq xyz$$

Paragraph for Question Nos. 29 to 30

Given the sequence of number $a_1, a_2, a_3 \ldots\ldots\ldots\ldots a_{1005}$

which satisfy $\frac{a_1}{a_1+1} = \frac{a_2}{a_2+3} = \frac{a_3}{a_3+5} = \ldots\ldots\ldots = \frac{a_{1005}}{a_{1005}+2009}$

Also $a_1 + a_2 + a_3 + \ldots\ldots\ldots\ldots + a_{1005} = 2010$

संख्याओं की श्रेणी a_1, a_2, a_3 ..... a_{1005} दी गई है।

जो कि $\frac{a_1}{a_1+1}=\frac{a_2}{a_2+3}=\frac{a_3}{a_3+5}=\ldots\ldots\ldots=\frac{a_{1005}}{a_{1005}+2009}$ तथा

$a_1 + a_2 + a_3 + \ldots\ldots\ldots\ldots + a_{1005} = 2010$ को संतुष्ट करती है।

29. Nature of the sequence is :

श्रेणी की प्रकृति होगी :

(A) A.P. (B) G.P. (C) A.G.P. (D) H.P.

Ans. A

Sol. $a_1, a_2, a_3, \ldots\ldots\ldots, a_{1005}$

$$\frac{a_1}{a_1+1}=\frac{a_2}{a_2+3}=\frac{a_3}{a_3+5}=\ldots\ldots\ldots=\frac{a_{1005}}{a_{1005}+2009}$$

$a_1 + a_2 + a_3 + \ldots\ldots\ldots + a_{1005} = 2010$

$a_1(a_2 + 3) = a_2(a_1 + 1) \Rightarrow \frac{a_2}{a_1} = 3 \Rightarrow a_2 = 3a_1$

$a_2(a_3 + 5) = a_3(a_2 + 3) \Rightarrow \frac{a_3}{a_2} = \frac{5}{3} \Rightarrow a_3 = 5a_1$

$a_3(a_4 + 7) = a_4(a_3 + 5) \Rightarrow \frac{a_4}{a_3} = \frac{7}{5} \Rightarrow a_4 = 7a_1$

$a_1 + a_2 + a_3 + \ldots\ldots\ldots + a_{1005} = 2010$

$a_1 + 3a_1 + 5a_1 + \ldots\ldots\ldots + 2009a_1 = 2010$

$a_1[1 + 3 + 5 \ldots\ldots\ldots + 2009] = 2010$

$$a_1\left[\frac{1005}{2}\left[2\times1+(1004)\times2\right]\right]=2010$$

$$a_1=\frac{2}{1005},\qquad d=2a_1=\frac{4}{1005}$$

Series form an A.P.

30. 21st term of the sequence is equal to :

श्रेणी का 21वां पद होगा :

(A) $\frac{86}{1005}$ (B) $\frac{83}{1005}$ (C) $\frac{82}{1005}$ (D) $\frac{79}{1005}$

Ans. C

Sol. $T_{21} = a_1 + 20d = a_1 + 40a_1 = 41a_1$

$$T_{21} = \frac{82}{1005}$$

Paragraph for Question Nos. 31 to 33

Let V_r denote the sum of the first 'r' terms of an arithmetic progression (A.P.) whose first term is 'r' and the common difference is (2r – 1).

Let $T_r = V_{r+1} - V_r - 2$ and $Q_r = T_{r+1} - T_r$ for r = 1, 2, ...

माना V_r एक समान्तर श्रेणी (A.P.) के प्रथम 'r' पदों के योग को प्रदर्शित करता है जिसका प्रथम पद 'r' है एवं सर्वान्तर (2r – 1) है।

माना $T_r = V_{r+1} - V_r - 2$ तथा $Q_r = T_{r+1} - T_r$ जहाँ r = 1, 2, ...

31. The sum $V_1 + V_2 + ... + V_n$ is :

$V_1 + V_2 + ... + V_n$ का योग है :

(A) $\frac{1}{12}n(n+1)(3n^2 - n + 1)$ (B) $\frac{1}{12}n(n+1)(3n^2 + n + 2)$

(C) $\frac{1}{2}n(2n^2 - n + 1)$ (D) $\frac{1}{3}(2n^3 - 2n + 3)$

Ans. B

Sol. $a_1 + a_2 + a_3 + ... + a_r = v_r$

$a_1 = r, d = 2r - 1$

$$v_r = \left(\frac{r}{2}\right)\left[2\times r + (r-1)(2r-1)\right]$$

$$= \frac{r}{2}\left[2r + 2r^2 - 3r + 1\right]$$

$$v_r = \frac{r}{2}\left[2r^2 - r + 1\right]$$

$$v_r = r^3 - \frac{r^2}{2} + \frac{r}{2}$$

$$v_1 + v_2 + ...+ v_n = \sum_{r=1}^{n}\left(r^3 - \frac{r^2}{2} + \frac{r}{2}\right)$$

$$= \left[\frac{n(n+1)}{2}\right]^2 - \frac{n(n+1)(2n+1)}{2\times 6} + \frac{n(n+1)}{2\times 2}$$

$$= \frac{n(n+1)}{2}\left[\frac{n(n+1)}{2} - \left(\frac{2n+1}{6}\right) + \frac{1}{2}\right]$$

$$= \frac{n(n+1)}{2}\left[\frac{3n(n+1)-(2n+1)-3}{6}\right]$$

$$= \frac{n}{12}(n+1)\left(3n^2+n+2\right)$$

32. T_r is always :

(A) an odd number (B) an even number (C) a prime number (D) a composite number

T_r हमेशा :

(A) एक विषम संख्या है (B) एक सम संख्या (C) एक अभाज्य संख्या (D) एक मिश्र संख्या

Ans. D

Sol. $T_r = v_{r+1} - v_r - 2$

$$= \left[(r+1)^3 - \frac{(r+1)^2}{2} + \left(\frac{r+1}{2}\right)\right] - \left[r^3 - \frac{r^2}{2} + \frac{r}{2}\right] - 2$$

$$= \left[(r+1)^2 - r^3\right] - \frac{1}{2}\left[(r+1)^2 - r^2\right] + \frac{1}{2}[r+1-r] - 2$$

$$= \left[(r+1)^2 + r^2 + r(r+1)\right] - \left[\frac{2r+1}{2}\right] + \frac{1}{2} - 2$$

$=[3r^2 + 3r + 1] - r - 2$
$= 3r^2 + 2r - 1$
$= (3r - 1)(r + 1)$
Composite number

33. Which one of the following is a correct statement?

(A) Q_1, Q_2, Q_3, ... are in A.P. with common difference 5
(B) Q_1, Q_2, Q_3, ... are in A.P. with common difference 6
(C) Q_1, Q_2, Q_3, ... are in A.P. with common difference 11
(D) $Q_1 = Q_2 = Q_3 =$ .....

निम्नलिखित में से कौनसा कथन सही है?

(A) Q_1, Q_2, Q_3, ... A.P. में है जिसका सर्वान्तर 5 है
(B) Q_1, Q_2, Q_3, ... A.P. में है जिसका सर्वान्तर 6 है
(C) Q_1, Q_2, Q_3, ... A.P. में है जिसका सर्वान्तर 11 है
(D) $Q_1 = Q_2 = Q_3 =$ .....

Ans. B

Sol. $a = r$ and $d = 2r - 1$

$V_r = \frac{r}{2}\,(2r + (r-1)\,(2r-1)] = \frac{r}{2}[2r + 2r^2 - 3r + 1] = \frac{r}{2}(2r^2 - r + 1)$

$V_{r+1} = \left(\frac{r+1}{2}\right)(2\,(r+1)^2 - (r+1) + 1) = \left(\frac{r+1}{2}\right)(2(r^2 + 2r + 1) - r)$

$= \left(\frac{r+1}{2}\right)(2r^2 + 3r + 2) = \frac{2r^3 + 5r^2 + 5r + 2}{2}$

$T_r = V_{r+1} - V_r - 2 = 3r^2 + 2r - 1 = 3r^2 + 3r - r - 1 = (3r-1)\,(r+1)$ (composite no.)

$\sum V_r = r^3 - \frac{r^2}{2} + \frac{r}{2} = \frac{n^2(n+1)^2}{4} - \frac{n(n+1)(2n+1)}{12} + \frac{n(n+1)}{4} = \frac{n(n+1)}{12}\,(3n^2 + 3n - 2n - 1 + 3)$

$= \frac{n(n+1)}{12}\,(3n^2 + n + 2)$

$Q_r = T_{r+1} - T_r = 3(r+1)^2 + 2(r+1) - 1 - 3r^2 - 2r + 1 = 3r^2 + 6r + 3 + 2r + 1 - 3r^2 - 2r + 1 = 6r + 5$

Common difference of Q_1, Q_2, Q_3 is 6 and they are in A.P.

Hindi. $a = r$ तथा $d = 2r - 1$

$V_r = \frac{r}{2}\,(2r + (r-1)\,(2r-1)] = \frac{r}{2}[2r + 2r^2 - 3r + 1] = \frac{r}{2}(2r^2 - r + 1)$

$V_{r+1} = \left(\frac{r+1}{2}\right)(2\,(r+1)^2 - (r+1) + 1) = \left(\frac{r+1}{2}\right)(2(r^2 + 2r + 1) - r)$

$= \left(\frac{r+1}{2}\right)(2r^2 + 3r + 2) = \frac{2r^3 + 5r^2 + 5r + 2}{2}$

$T_r = V_{r+1} - V_r - 2 = 3r^2 + 2r - 1 = 3r^2 + 3r - r - 1 = (3r-1)\,(r+1)$ (संयुक्त संख्या)

$\sum V_r = r^3 - \frac{r^2}{2} + \frac{r}{2} = \frac{n^2(n+1)^2}{4} - \frac{n(n+1)(2n+1)}{12} + \frac{n(n+1)}{4} = \frac{n(n+1)}{12}\,(3n^2 + 3n - 2n - 1 + 3)$

$= \frac{n(n+1)}{12}\,(3n^2 + n + 2)$

$Q_r = T_{r+1} - T_r = 3(r+1)^2 + 2(r+1) - 1 - 3r^2 - 2r + 1 = 3r^2 + 6r + 3 + 2r + 1 - 3r^2 - 2r + 1 = 6r + 5$

अतः Q_1, Q_2, Q_3 समान्तर श्रेढ़ी में हैं जिनका सार्वअन्तर 6 है

MATCH THE COLUMN TYPE QUESTION(S)

34. Natural numbers are divided into groups as follows {1}, {2, 3}, {4, 5, 6}, {7, 8, 9, 10} …….. Now match the following :

Column - I	**Column - II**
(A) Sum of numbers in 100th group	(P) 125500
(B) Middle number in 501th group	(Q) 500050
(C) Last number in 1000th group	(R) 319601
(D) First number of 800th group	(S) 500500

प्राकृतिक संख्याओं का एक समूह निम्न प्रकार से विभक्त किया गया है, {1}, {2, 3}, {4, 5, 6}, {7, 8, 9, 10} …….. तो निम्न का मिलान कीजिये :

Column - I	**Column - II**
(A) 100वें समूह में संख्याओं का योगफल है	(P) 125500
(B) 501वें समूह में मध्य वाली संख्या है	(Q) 500050
(C) 1000वें समूह में अन्तिम संख्या है	(R) 319601
(D) 800वें समूह में प्रथम संख्या है	(S) 500500

Ans. A-Q; B-P;C-S;D-R

Sol. {1}, {2, 3}, {4, 5, 6}, {7,8,9,10}, ...

Number of digits in 100th group

$t_{100} = 1 + (100 - 1) \times 1$

$t_{100} = 100$

First term of 100th group

$S = 1 + 2 + 4 + 7 + ... + T_{n-1} + T_n$

$S = \quad 1 + 2 + 4 + ... + T_{n-2} + T_{n-1} + T_n$

$$0 = 1 + \underbrace{(1 + 2 + 3 + ...)}_{(n-1) \text{ term}} - T_n$$

$$T_n = 1 + \frac{(n-1)n}{2} = \frac{2 + n(n-1)}{2}$$

$$T_{100} = \frac{2 + 100 \times 99}{2} = 4951$$

Sum of numbes in 100th group $= \frac{100}{2}[2 \times 4951 + 99 \times 1] = 500050$

(B) First tem of 501th goup $= \frac{2 + (501) \times 500}{2} = 125251$

middle term of 501th group = 125251 + 249 = 125500

(C) First term of 1000th goup $= \frac{2 + 1000 \times 999}{2} = 499501$

Last term of 1000th goup = 499501 + 999 = 500501

(D) First number of 800th group $= \frac{2+800\times 799}{2} = 319601$

35. Match the following columns :

Column I	**Column II**
(A) The value of $(2^{1/4}\, 4^{1/8}\, 8^{1/16}\, 16^{1/32} \ldots)$ is	(P) 1
(B) S_n denote the sum of the first n terms of an A. P. If $S_{2n} = 3S_n$, then the value of S_{3n}/S_n is	(Q) 2
(C) The value of $(0.2)^{\log_{\sqrt{5}}\left(\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\ldots\right)}$ is	(R) 4
(D) If H is H. M. between p and q, then $\frac{H}{p}+\frac{H}{q} =$	(S) 6

निम्न Columns को सुमेलित कीजिए :

Column I	**Column II**
(A) $(2^{1/4}\, 4^{1/8}\, 8^{1/16}\, 16^{1/32} \ldots)$ का मान है	(P) 1
(B) S_n किसी स.श्रे. के प्रथम n पदों के योगफल को व्यक्त करता है यदि $S_{2n} = 3S_n$, तो S_{3n}/S_n का मान है	(Q) 2
(C) $(0.2)^{\log_{\sqrt{5}}\left(\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\ldots\right)}$ का मान है	(R) 4
(D) यदि H संख्याओं p तथा q के मध्य ह.माध्य है तो $\frac{H}{p}+\frac{H}{q} =$	(S) 6

Ans. A - Q, B - S, C - R, D - Q

Sol. (A) $2^{\frac{1}{4}}\cdot 2^{\frac{2}{8}}\cdot 2^{\frac{3}{16}}\cdot 2^{\frac{4}{32}}\ldots\ldots$

$$S = 2^{\left(\frac{1}{4}+\frac{2}{8}+\frac{3}{16}+\frac{4}{32}+\ldots\ldots\right)}$$

$$S' = \frac{1}{2^2}+\frac{2}{2^3}+\frac{3}{2^4}+\frac{4}{2^5}+\ldots\ldots\ldots \quad ...(1)$$

$$\frac{S'}{2} = 0+\frac{1}{2^3}+\frac{2}{2^4}+\frac{3}{2^5}+\ldots\ldots\ldots \quad ...(2)$$

$$\frac{S'}{2} = \frac{1}{2^2}+\frac{1}{2^3}+\frac{1}{2^4}+\frac{1}{2^5}+\ldots\ldots\ldots = \frac{\frac{1}{2^2}}{1-\frac{1}{2}} = \frac{1}{2}$$

(B) $S_n = \frac{n}{2}\left[2a+(n-1)d\right]$

$$S_{2n} = \frac{2n}{2}\left[2a + (2n-1)d\right] = 3 \times \frac{n}{2}\left[2a + (n-1)d\right]$$

$4a + (4n - 2)d = 6a + (3n - 3)d$

$-2a + (n + 1)d = 0$

$2a = +(n + 1)\, d$ ...(i)

$$\frac{S_{3n}}{S_n} = \frac{\frac{3n}{2}\left[2a + (3n-1)d\right]}{\frac{n}{2}\left[2a + (n-1)d\right]} = \frac{3\left[+nd + d + +3nd - d\right]}{\left[+nd + d + nd - d\right]}$$

$$= 3\left[\frac{4nd}{2nd}\right] = 6$$

(C) $(0.2)^{\log_{\sqrt{5}}\left(\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+.....\right)}$

$$= (0.2)^{2\log_5\left(\frac{\frac{1}{4}}{1-\frac{1}{2}}\right)}$$

$$= (0.2)^{2\log_5\frac{1}{2}}$$

$$= \left(\frac{1}{5}\right)^{-2\log_5 2}$$

$$= 5^{2\log_5 2} = 5^{\log_5 2^2} = 2^2 = 4$$

(D) $H = \dfrac{2pq}{p+q}$

$$H\left[\frac{1}{p} + \frac{1}{q}\right] = H\left[\frac{p+q}{pq}\right] = 2$$